Semester V **B. Tech CSE (AI)**

BCSAI 507: DATABASE SECURITY

2L + 0T + 0P + 2C **MM 100**

Unit 1: Security Architecture & Operating System Security Fundamentals

Security Architecture: Introduction-Information Systems- Database Management Systems-Information Security Architecture- Database Security–Asset Types and value-Security Methods. Operating System Security Fundamentals: Introduction-Operating System Overview-Security Environment – Components-Authentication Methods-User Administration-Password PoliciesVulnerabilities-E-mail Security

Unit 2 : Administration Of Users & Profiles,Password Policies, Privileges And Roles

Administration of Users: Introduction-Authentication-Creating Users, SQL Server User-Removing, Modifying Users-Default, Remote Users-Database Links-Linked Servers-Remote Servers-Practices for Administrators and Managers-Best Practices Profiles, Password Policies, Privileges and Roles: Introduction-Defining and Using Profiles-Designing and Implementing Password Policies-Granting and Revoking User Privileges-Creating, Assigning and Revoking User Roles-Best Practices

Unit 3: Database Application Security Models & Virtual Private Databases

Database Application Security Models: Introduction-Types of Users-Security Models- Application Types-Application Security Models-Data Encryption Virtual Private Databases: Introduction-Overview of VPD-Implementation of VPD using Views, Application Context in Oracle-Implementing Oracle VPD- Viewing VPD Policies and Application contexts using Data Dictionary, Policy Manager Implementing Row and Column level Security with SQL Server

Unit 4: Auditing Database Activities

Auditing Database Activities: Using Oracle Database Activities-Creating DLL Triggers with OracleAuditing Database Activities with Oracle-Auditing Server Activity with SQL Server 2000-Security and Auditing Project Case Study.

Unit 5: Privacy Preserving Data Mining Techniques

Privacy Preserving Data Mining Techniques: Introduction- Privacy Preserving Data Mining AlgorithmsGeneral Survey-Randomization Methods-Group Based Anonymization-Distributed Privacy Preserving Data Mining-Curse of Dimensionality-Application of Privacy Preserving Data Mining

Text/Reference Books:

1. Hassan A. Afyouni, “Database Security and Auditing”, Third Edition, Cengage Learning,2009
2. 2.Charu C. Aggarwal, Philip S Yu, “Privacy Preserving Data Mining”: Models and Algorithms, Kluwer Academic Publishers, 2008
3. Ron Ben Natan, ”Implementing Database Security and Auditing”, Elsevier Digital Press, 2005.
1. http://charuaggarwal.net/toc.pdf